

Assignment 4 - Full Stack Application

(MySQL + ORM, WebSockets, and AI Integration)

Overview

In this assignment you will upgrade your project into a complete full-stack application. By the end, your application must include:

- React frontend
- Node.js + Express backend
- MySQL database
- ORM-based database mapping
- WebSocket integration
- AI integration

Your goal is to finish with an application that stores data persistently, supports real-time communication, and includes at least one AI-powered feature.

Your goal is to finish with an application that:

- starts and runs locally.
- stores data persistently, which must persist after server restarts.
- supports real-time communication.
- includes at least one AI-powered feature relevant to the project domain.

Learning Goals

- Design relational database schemas.
- Use ORM models and relationships.
- Replace mock data with a real database.
- Implement real-time communication using Socket.IO.
- Integrate AI services into a web application.
- Connect frontend, backend, database, WebSockets, and AI into one system.

Technical Requirements

Frontend:

- React.js
- React Router
- Fetch API or Axios

Backend:

- Node.js
- Express

Database:

- MySQL

ORM:

- Any ORM framework that answer this assignment requirements will be accepted.

WebSockets:

- Socket.IO

AI:

- Any AI model, interface or API of your choice.

Part 1 - MySQL + ORM Integration

Replace all mock data from previous assignments with a real MySQL database.

Required Models:

- User
- Admin
- Main project resource (depending on your idea)
- Junction Table

Required Features:

- Create records
- Read records
- Update records
- Delete records
- Relational queries (JOIN)

Required Relationships:

Implement at least two ORM relationships, one-to-many AND many-to-many.

All data must persist after server restarts.

Part 2 - WebSocket Integration

Implement one meaningful real-time feature.

Examples:

- Real-time chat
- Live notifications
- Online user status
- Live dashboard updates

Requirements:

- Use Socket.IO
- Implement at least three custom events (IN ADDITION TO “connect” and “disconnect”)
- Demonstrate communication between at least two browser tabs or clients
- Integrate the feature into the frontend UI

Part 3 - AI Integration

Implement one meaningful AI-powered feature.

Examples:

- Summarization
- Recommendation system
- Content generation
- Classification
- AI assistant

Requirements:

- Expose at least one AI API endpoint
- Frontend must communicate through the backend
- AI provider keys must not be exposed to the frontend
- AI functionality must be relevant to the project domain

API Requirements

All API responses must follow the Assignment 2 format.

Success:

```
{
  "success": true,
  "data": {},
  "error": null
}
```

Error:

```
{
  "success": false,
  "data": null,
  "error": {
    "code": "ERROR_CODE",
    "message": "Human readable message",
    "details": {}
  }
}
```

Required project structure

Organize your Full Stack project following this structure:

- **frontend/src/** - React source code (Components, Pages, Services, App.js).
- **backend/src/** - Node.js + Express backend source code.
- **backend/models/** - ORM models and mapping files.
- **backend/migrations/** - Database migrations or schema files.

Documentation Requirements

Create a `README.md` that explains:

- Project purpose
- Installation instructions
- Database setup
- Environment variables
- ORM setup
- API endpoints
- WebSocket feature
- AI feature
- Known limitations

The documentation should allow a classmate to run the project without asking questions.

What You Must Submit

Submit a ZIP containing:

- Frontend source code
- Backend source code
- ORM models
- Migrations/schema files
- `README.md`
- `.env.example`
- Postman collection
- **Screenshots Required:** Include screenshots showing:
 - Database-connected application
 - Successful CRUD operation
 - ORM relationships working
 - WebSocket communication between two clients
 - AI feature input and output
 - Database tables or migrations
- Demo video

Do not submit:

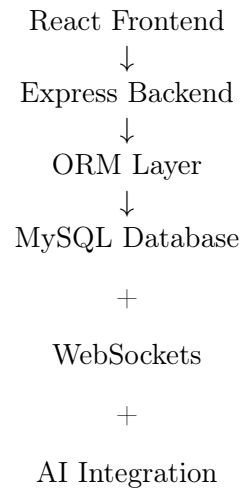
- `node_modules`
- Real API keys
- Real passwords
- `.env` files containing secrets

Evaluation Criteria

- MySQL + ORM Integration: 30%
- WebSocket Integration: 30%
- AI Integration: 20%
- Frontend Integration: 10%
- Documentation Quality: 10%

Summary

By the end of this assignment you should have a complete full-stack application consisting of:



This assignment prepares your project for final deployment and presentation.